Day-10

APTITUDE

1) A car travels 60km in 1 hour. what is the speed?

Sol...

D = 60km

T = 1hr

speed = D/T

⇒ 60/1 ⇒ 60km/hr

$\boxed{60km/hr}$

2) A train travels at 50km/hr. How long will it take to travel 150km?

Sol:- speed = 50km/hr

D = 150km

T = D/S

$\Rightarrow \frac{\cancel{150}^{3}}{\cancel{50}}$

$\boxed{\Rightarrow 3 \text{ hours}}$

3) If a person walks at 5km/hr, how far will they walk in 3 hours?

speed = 5km/hr

T = 3 hours

D = speed × time

⇒ 5 × 3

⇒ 15 km

$\boxed{15km}$

4) Two trains moves in opposite direction at 40km/hr & 60km/hr. Relative speed.

Sol:-

Two trains in opposite direction → (+)

Relative speed = sum of speeds

⇒ 40+60km/hr

⇒ 100km/hr

$\boxed{100 km/hr}$

5) A car travels first half distance at 40km/hr and second half at 60km/hr. Find average speed.

car1 = half of 40km/hr

car2 = half of 60km/hr

Average speed → $(2 \times S1 \times S2) / (S_1 + S_2)$

$$V_{avg} = \frac{2V_1V_2}{V_1+V_2}$$

$V_1 = 40 \Rightarrow V_2 = 60$.

$$\rightarrow \frac{(2 \times 40 \times 60)}{40+60}$$

$$\Rightarrow \frac{48\cancel{00}}{1\cancel{00}}$$

⇒ 48km/hr

$\boxed{48 km/hr}$

Block 2 - VERBAL ABILITY

1) where does binary search work?

Binary search works on sorted arrays.

2) How does it reduce search space?

It repeatedly divides the search interval into half and reduces the search space.

3) Is binary search faster than linear search?

Yes binary search faster than linear search.

Grammar Correction:

1) He don't know binary search.

He doesn't know binary search

2) She have completed the task.

She has completed the task.

3) They is learning algorithm.

They are learning algorithm.

Block 3 - COMMUNICATION PRACTICE

Explain Binary search algorithms.

Binary search is an efficient algorithm. It is work on sorted arrays.

It repeatedly divides the search interval into half and reduces the search space.

First, we find the middle element of the array. If the middle element is equal to the target value, the search is successful. If the ~~element~~ target value is smaller than middle element, we search in left half of the array. If the target value is greater, we search in the right half. ~~until do~~ we ~~this step~~ repeatedly ~~to find~~ repeat the step until the target value is found.

The time complexity of binary search is $O(\log n)$.